12 kW Wind Turbine for Iceland: Smaller Generator Drum (Ø490 and Ø290) with 30 × 15 × 12 Magnets

Status: **study** · 2026-11-05 (Martin: “make a version with a drum diameter less then 500mm and less then 300mm, using 30x15x12 blocks and make a document comparing the larger version (with same magnets) also how it will affect the energy output”)
Builds on: 06 – Generator round 2 (electromagnetic design, losses), 12 – Power electronics (efficiency chain, annual energy), 20 – N42SH 30 × 15 × 6 variant (the magnet pole used here)
Analysis: `analysis/generator/smalldrum.py` → `results/small_P1.json`, `small_D490.json`, `small_D290.json`, `small_summary.json`

Summary

The turbine rotor turns at 178 rpm at rated wind whatever the generator looks like. A direct-drive generator’s power therefore comes only from its torque. Torque grows with the **square of the air gap diameter** times the stack length, and a smaller machine also has room for less copper. Halving the drum diameter cuts the torque of the same stack length by much more than half.

	Present (doc 20)	Ø490, full power	Ø490, 74 mm stack	Ø290, full power	Ø290, 74 mm stack
Drum outside diameter	646 mm	490 mm	490 mm	290 mm	290 mm
Poles / slots	40 / 45	32 / 36	32 / 36	16 / 18	16 / 18
Stack length	74 mm	**155 mm**	74 mm	**1,128 mm**	74 mm
Torque at sizing current	802 Nm	802 Nm	383 Nm	802 Nm	53 Nm
Rated shaft power	13.6 kW	13.6 kW	**6.5 kW**	13.6 kW	**0.9 kW**
Generator efficiency at rated	93.9 %	93.8 %	92.9 %	91.7 %	84.7 %
30 × 15 × 6 blocks (magnet mass)	400 (8.2 kg)	640 (13.1 kg)	320 (6.6 kg)	2,368 (48.5 kg)	160 (3.3 kg)
Active mass (magnet, copper, laminations, drum band)	89 kg	122 kg	61 kg	326 kg	23 kg
Annual energy at 6 m/s average	31.6 MWh	31.5 MWh (−0.1 %)	22.1 MWh (**−30 %**)	30.9 MWh (−2 %)	3.4 MWh (**−89 %**)
Annual energy at 8 m/s average	51.1 MWh	51.0 MWh (−0.1 %)	30.9 MWh (**−40 %**)	50.0 MWh (−2 %)	4.0 MWh (**−92 %**)

What it means:

- **Under 500 mm (Ø490):** this is **feasible if the stack grows from 74 to 155 mm**. Energy output is unchanged (−0.1 %). The cost is 60 % more magnet (640 blocks), 40 % more

laminations, and a drum 81 mm longer, which has to fit the nacelle and spindle layout. Keeping the 74 mm stack halves the generator: 6.5 kW rated, and 22 to 44 % less energy per year depending on the site.

- **Under 300 mm (Ø290):** this is **not workable as a direct-drive 12 kW generator**. With a 74 mm stack it is a 0.9 kW generator, about 90 % less energy. For full power it would need a 1.13 m long stack with 48 kg of magnet and 326 kg of active material, almost four times the present machine. The Ø140 main spindle also leaves only 6.5 mm for the stator carrier wall, so the slots can only be 30 mm deep. A generator this small only makes sense with a gearbox (about 1:15, so the 74 mm Ø290 machine runs at about 2,700 rpm), which is a different turbine concept from the direct drive chosen in doc 04.
- **Same magnets, same pole:** both small machines keep the doc 20 pole (30 × 15 × 12, about the same pole pitch, rows skewed). Their short-circuit margin at 100 °C is 26 to 27 %, against 32 % in the present machine, because the smaller machines' fault current is relatively higher. Both are still above the 20 % rule at 100 °C.

Recommendation: if the present Ø646 drum is too big for the nacelle or for transport, **Ø490 with a 155 mm stack** is the realistic smaller generator. It gives the same energy, but it is heavier and longer and needs a nacelle fit check. A drum under 300 mm cannot give 12 kW in direct drive.

1. How the small machines were laid out

Every machine uses the doc 20 rotor pole:

- one 30 mm wide, 12 mm magnet across each pole (two N42SH 30 × 15 × 6 blocks bonded);
- rows of 15 mm blocks along the stack;
- 0.15 bond, 2 mm air gap, 0.5 mm sleeve;
- 9 mm of drum steel under the seat.

The stator keeps:

- the P1's tooth-coil winding family (9 slots per 8 poles);
- the same share of tooth width in the slot pitch;
- the same current density and slot fill (doc 06 sizing point).

	P1 (present)	Ø490	Ø290
Air gap diameter	600	444	243
Drum seat	Ø628.0	Ø472.0	Ø272.0
Poles / slots	40 / 45	32 / 36	16 / 18
Pole pitch at the magnet / pole span	47.4 / 0.63	43.9 / 0.68	48.3 / 0.62
Electrical frequency at 178 rpm	59.3 Hz	47.5 Hz	23.7 Hz
Slot depth	65	60	**30** (spindle)
Stator bore	Ø437	Ø292	**Ø153** (spindle Ø140)
Copper area per slot side	466 mm²	362 mm²	194 mm²

For the Ø290 machine the slots are limited twice. They close up towards the small inner radius, and the Ø140 main spindle (doc 07) has to pass through the stator. That leaves 30 mm deep slots, so the Ø290 machine has only 41 % of the P1's ampere-turns per metre of circumference on top of its smaller diameter.

2. FEA results (per 74 mm of stack, magnets at 100 °C)

	P1	Ø490	Ø290
Torque at the sizing current density	760 Nm (2D)	364 Nm	50 Nm
Torque relative to P1	1	0.48	0.066
Expected from D^2 × electric loading	1	0.46	0.067
Line EMF per turn per coil, 178 rpm	13.1 V	8.0 V	2.0 V
Turns per coil for 380 V (74 mm stack)	29	47	186
Turns per coil for 380 V (full-power stack)	29	23	12
Short-circuit current / sizing current	1.9	2.4	5.3
Transient short-circuit margin at 100 °C	32 %	26 %	27 %

The FEA confirms the simple rule: torque goes with D^2 times electric loading. The 2D torques are scaled by the same factor, so that the P1 makes its doc 06 value of 802 Nm. End effects are left out, which slightly flatters the short Ø290 stack.

3. Two ways to use a smaller drum

A. Keep 12 kW: lengthen the stack. The stack length needed is 802 Nm divided by the torque per metre:

- Ø490: 155 mm, which is 10 rows of magnets instead of 5;
- Ø290: 1,128 mm.

Copper loss stays about the same at Ø490 (503 W against 496 W). The end turns are a smaller share of the longer coils, which balances the longer stack. At Ø290 copper loss rises to 803 W, and iron loss stays about the same (more laminations, lower frequency).

B. Keep the 74 mm stack: the generator caps the power. The torque at the same current density is the new limit. The turbine pitches to hold that power, so rated power falls to 6.5 kW (Ø490) or 0.9 kW (Ø290).

4. Energy output

The doc 12 power curve and efficiency chain is used: rotor optimum below rated, pitch-limited above, active rectifier, grid inverter, 120 W auxiliaries, Rayleigh wind and 97 % availability. Only the generator’s losses and torque limit change. Net energy per year:

Average wind speed	5 m/s	6 m/s	7 m/s	8 m/s	10 m/s
Present Ø646, 74 mm (MWh)	20.7	31.6	42.0	51.1	65.1
Ø490, 155 mm	20.7 (−0.2 %)	31.5 (−0.1 %)	41.9 (−0.1 %)	51.0 (−0.1 %)	65.0 (−0.1 %)
Ø490, 74 mm	16.2 (−22 %)	22.1 (−30 %)	27.0 (−36 %)	30.9 (−40 %)	36.2 (−44 %)
Ø290, 1,128 mm	20.3 (−2.0 %)	30.9 (−2.1 %)	41.0 (−2.2 %)	50.0 (−2.3 %)	63.6 (−2.3 %)
Ø290, 74 mm	2.9 (−86 %)	3.4 (−89 %)	3.7 (−91 %)	4.0 (−92 %)	4.3 (−93 %)

The windier the site, the more a power-capped generator loses, because more of the year is spent above its smaller rated power. Iceland’s coastal and highland sites (7 to 10 m/s) are where the 74 mm Ø490 version loses most: 36 to 44 %.

5. What else changes

	Ø490, 155 mm	Ø290, 1,128 mm
Drum	81 mm longer, Ø490; land and comb bars for 10 rows	1.1 m long tube; not practical on the hub
Spindle and bearings (doc 07)	Stator carrier longer; check that the bearing spacing and the brake disc still fit	Spindle must carry a 1.1 m stator; new layout
Winding	23 turns per coil, 36 coils	12 turns per coil, 18 coils
Cooling	Copper loss about the same, gap area 2.2 × larger: cooler per m²	1.6 × the present copper loss (803 W), through a Ø153 bore with no room for an air channel
Power electronics	Unchanged (same voltage and current class)	Unchanged electrically; 24 Hz at rated
Magnets	640 + 32 blocks, 13.1 kg	2,368 blocks, 48.5 kg
Short-circuit margin at 100 °C	26 % (above the 20 % rule)	27 %

6. Open items

#	Item
O1	If Ø490 is wanted: nacelle and spindle fit check for a 155 mm stack and a drum 81 mm longer (docs 07 and 16), and drum stiffness for 10 rows
O2	What drives the smaller drum: transport size, nacelle cover, weight or cost? With that known, a middle size (for example Ø560 with about a 100 mm stack) may be the better compromise
O3	Full doc 06 thermal check of the chosen size

Appendix. How to rerun

```
cd analysis/generator
python3 smalldrum.py fea P1 D490 D290    # FEA, ~5 min each (P1 first) ->
results/small_<key>.json
python3 smalldrum.py energy              # losses, power curves and annual energy ->
results/small_summary.json
```